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Software Note

'biomod2' – extending presence–absence species distribution models to multiple data types

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The R package 'biomod2' is one of the most widely used and versatile tools for species distribution modelling (SDM), enabling ecologists to calibrate, evaluate, and project species–environment relationships across space and time using multiple modelling algorithms and ensemble forecasting. Here, we present its latest and most comprehensive version, which substantially expands its scope by accommodating diverse ecological data types within a unified modelling workflow. Importantly, 'biomod2' is designed to handle each data type (e.g. presence-only, presence–absence, counts, multi-class abundance, or relative/absolute abundance) independently, ensuring methodological rigor and avoiding the integration of incompatible data formats or model types. This approach allows users to model biomass, land cover, habitat suitability, and other ecological metrics without conflating distinct data streams or analytical frameworks.

We have fully restructured the workflow to improve usability and reproducibility, introducing clearer function organization, standardized parameter names, and substantially revised documentation. The update also includes new methodological features such as enhanced cross-validation schemes, improved pseudo-absence selection strategies, expanded model parametrization options, additional algorithms, and new tools for exploring and visualizing outputs. Together, these developments provide a more flexible, transparent, and easily shareable framework for SDM, supporting comparison, uncertainty analyses, and ensemble predictions across a broad range of ecological questions.

Keywords: abundance, habitat distribution model, reproducibility, species distribution model

Introduction

The importance of species distribution models (SDMs) in ecology is well established, and the number of publications employing SDMs has grown steadily since the early



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2000s. Ensemble modelling, which allows running multiple algorithms on multiple subsets of data, in particular, has seen a dramatic expansion: a query for ‘species distribution models’ and ‘ensemble models’ on the OpenAlex platform returns over 3300 articles, representing an eightfold increase from 39 articles in 2010 to 335 in 2024. Ensemble approaches are now frequently integrated into conservation planning (Guisan et al. 2013, Meller et al. 2014, Lin et al. 2018), owing to their capacity to run multiple algorithms, to compare them, to combine certain predictions if needed, and potentially increase predictive performance while better assessing uncertainty (Araújo and New 2007, Thuiller et al. 2019).

Since its inception, ‘BIOMOD’ (Thuiller 2003, Thuiller et al. 2009), now known as ‘biomod2’ (Thuiller 2014), has provided ecologists with a comprehensive framework for generating, evaluating, and comparing SDM outputs within R. Given the widespread adoption of ‘biomod2’, we have undertaken substantial efforts since 2022 to modernize

and extend the package. These efforts include refactoring and organizing the code base, standardizing parameter nomenclature, enhancing documentation (with detailed examples), and establishing an online platform for distribution and support (<https://biomodhub.github.io/biomod2/>). Multiple releases have followed, each enriching the package: support was migrated from the ‘raster’ to the ‘terra’ package for faster and more memory-efficient spatial data handling; cross-validation routines and pseudo-absence selections were expanded; and novel algorithms such as deep neural networks (DNN) or XGBoost were recently integrated (Fig. 1).

In 2025, responding to the growing demand for modelling non-binary data, we have extended ‘biomod2’ to accommodate abundance-related data types. This represents a significant departure from the package’s historical focus on presence–absence or presence-only data (augmented by pseudo-absences), enabling users to make full use of the diversity of available ecological observations when constructing SDMs.

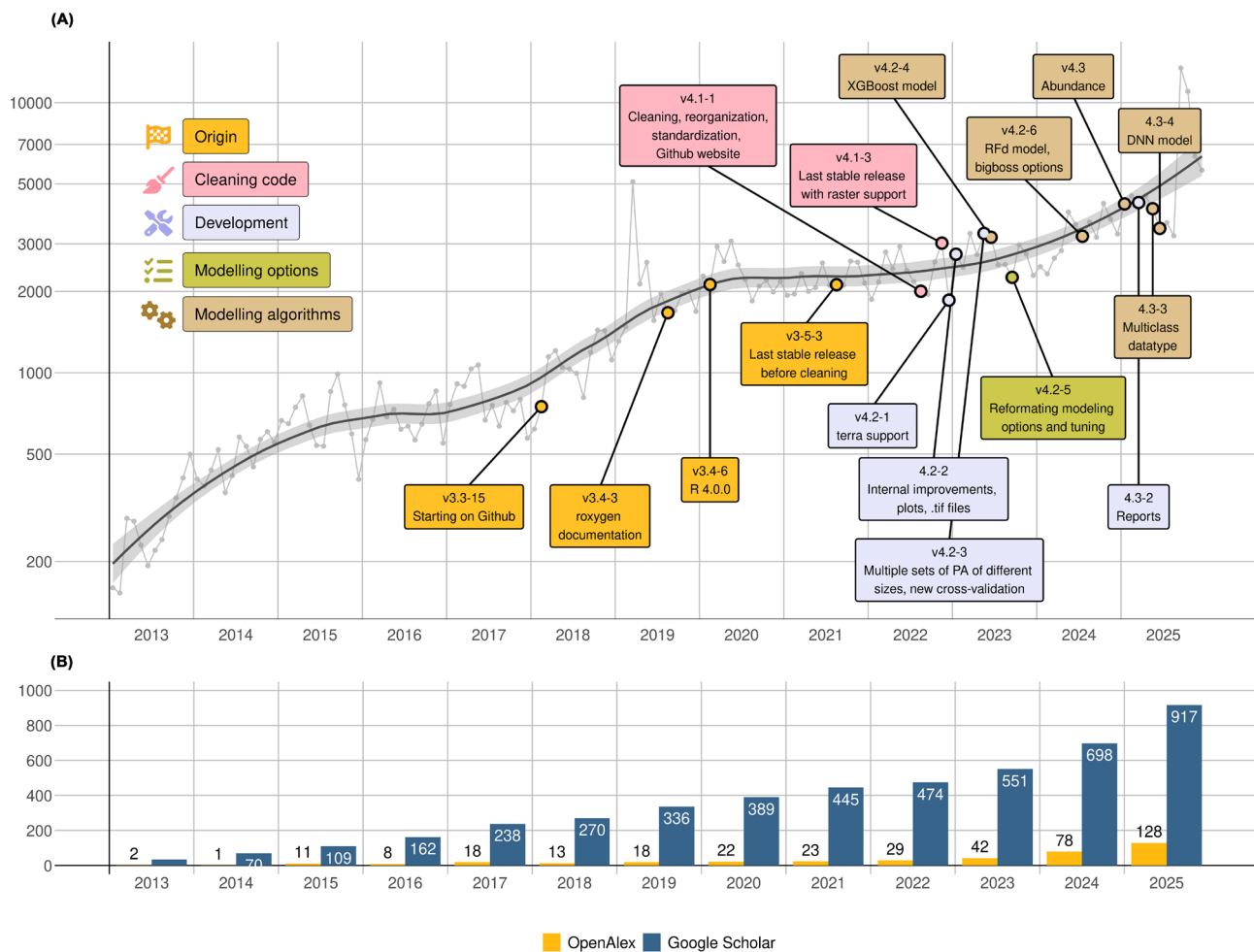


Figure 1. (A) Evolution through time of the number of monthly downloads of ‘biomod2’ package from the RStudio CRAN mirror (obtained with the *cran_downloads* function from the ‘cranlogs’ package); important version updates are highlighted. These download numbers do not take into account direct package installation from the GitHub repository. (B) Evolution through time of the number of yearly results searching for the term ‘biomod2’ on different scholar platforms.

Despite reviews and comparative studies of abundance-based SDMs (Pearce and Boyce 2006, Baldrige et al. 2016, Waldock et al. 2022), few unified workflows existed so far (de Oliveira Junior and Velazco 2025). This lack of reproductive framework mostly stemmed from two issues. First, abundance-related data (e.g. abundance, biomass, land cover proportion) were so far not widely available compared to the ever-increasing availability of presence-only data (e.g. through citizen-science). Yet, significant progress is being made regarding interpretation of non-binary data from novel technologies such as camera-traps, sound recorders, and environmental DNA metabarcoding, into count/frequency/abundance-related estimations (Moxley et al. 2017, Moeller et al. 2018, Pochardt et al. 2020, Hartig et al. 2024). Second, abundance (or count estimates) is notoriously difficult to model (Guisan and Harrell 2000, Forney et al. 2012, Becker et al. 2019, Gao et al. 2025) because its spatial variation is not only determined by environmental gradients but also by demographic stochasticity, mass effects, and biotic interactions, which are rarely explicitly accounted for (Boulangeat et al. 2012). Addressing this difficulty calls for a flexible framework able to test multiple algorithms (parametric versus non-parametrics, linear versus non-linear, regression versus tree-based), quantify uncertainty, and deliver optimized predictions, a need the new version of 'biomod2' is designed to meet.

Compared to ver. 3.5-1, the new release (4.3-4) introduces three major internal changes:

1. Clear separation of main and auxiliary functions (Supporting information). Main functions (prefixed *BIOMOD_* [...]) implement the core workflow: data formatting, modelling, and projections. They take as input and return 'biomod2' objects. Auxiliary functions (prefixed *bm_* [...]) perform lower-level tasks: pseudo-absences and cross-validation sampling, modelling options setting, and plotting routines, and can be invoked independently if desired. Plotting routines also take 'biomod2' objects as input and generate preprocessed graphical outputs. Getter functions have been standardized to facilitate consistent access to intermediate results.
2. Centralized management of single-model algorithms. A new *ModelTable* data structure enumerates all supported algorithms, along with their requisite R packages and function calls (Supporting information). This design allows 'biomod2' to retrieve default parameter settings automatically, offer flexible tuning of modelling options, and standardize how individual algorithms are executed within the ensemble framework.
3. Incorporation of a *data.type* argument for richer data support. Beyond presence-absence and presence-only (with pseudo-absences), 'biomod2' ver. 4.3-4 now supports counts, multiclass, ordinal, relative, and absolute abundance values. This parameter is propagated through both the formatted input data and the model-specific settings, as nearly all single-model algorithms have been re-parameterized accordingly (Supporting information).

In this manuscript, we describe 'biomod2' ver. 4.3-4, detailing its revised and novel functionalities through an illustrative example involving abundance data. The stable release is available on CRAN, while the development ver. (4.3-4-6 on March 2026) can be accessed on GitHub (<https://github.com/biomodhub/biomod2/releases/tag/v4.3-4-6>). Given the extensive use of 'biomod2' in the literature, earlier versions have been the subject of numerous reviews. Here, we highlight the package's previous limitations, describe the improvements introduced in ver. 4.3-4, and discuss broader considerations for species distribution modelling.

'biomod2' package

With a concise set of functions, 'biomod2' enables the fitting of over a dozen different algorithms for species distribution modelling, which can subsequently be combined into one or more ensemble models. Users can generate projections at both the single-model and ensemble levels, and a suite of utilities facilitate diverse pseudo-absence strategies, cross-validation schemes, modelling options, and evaluation metrics. The complete workflow is outlined in Fig. 2 and the Supporting information.

Data formatting

All necessary input data, such as species identifier, data values and data type, geographic coordinates, and environmental predictors (continuous and/or categorical), are compiled through the *BIOMOD_FormatingData* function. If independent evaluation data are available, these can likewise be integrated for model validation.

Data type specification

The *data.type* argument supports six distinct categories, which inform both data preparation and compatible algorithms: *binary* (values: NA, 0, 1), *count* (non-negative integers), *multiclass* (categorical classes), *ordinal* (ordered categorical classes), *relative* (continuous values in [0, 1]), and *abundance* (positive continuous values). 'biomod2' conditionally adjusts downstream routines such as pseudo-absence sampling, algorithm selection, and appropriate evaluation metrics, to match the nature of the response variable.

Pseudo-absence sampling (*bm_PseudoAbsences*)

An important point is that 'biomod2' generates pseudo-absence points and not background points. In species distribution modelling, the distinction is subtle but important (Renner et al. 2015). Pseudo-absences are points sampled from locations where the species was not recorded, and are treated as true absences in the model. They are used when only presence data are available, to provide a contrast against which the model can learn what environments the species avoids or does not occupy. Their selection is often constrained, for instance, by excluding areas too close to known presences, or by targeting environmentally distinct regions, because they are meant to approximate real

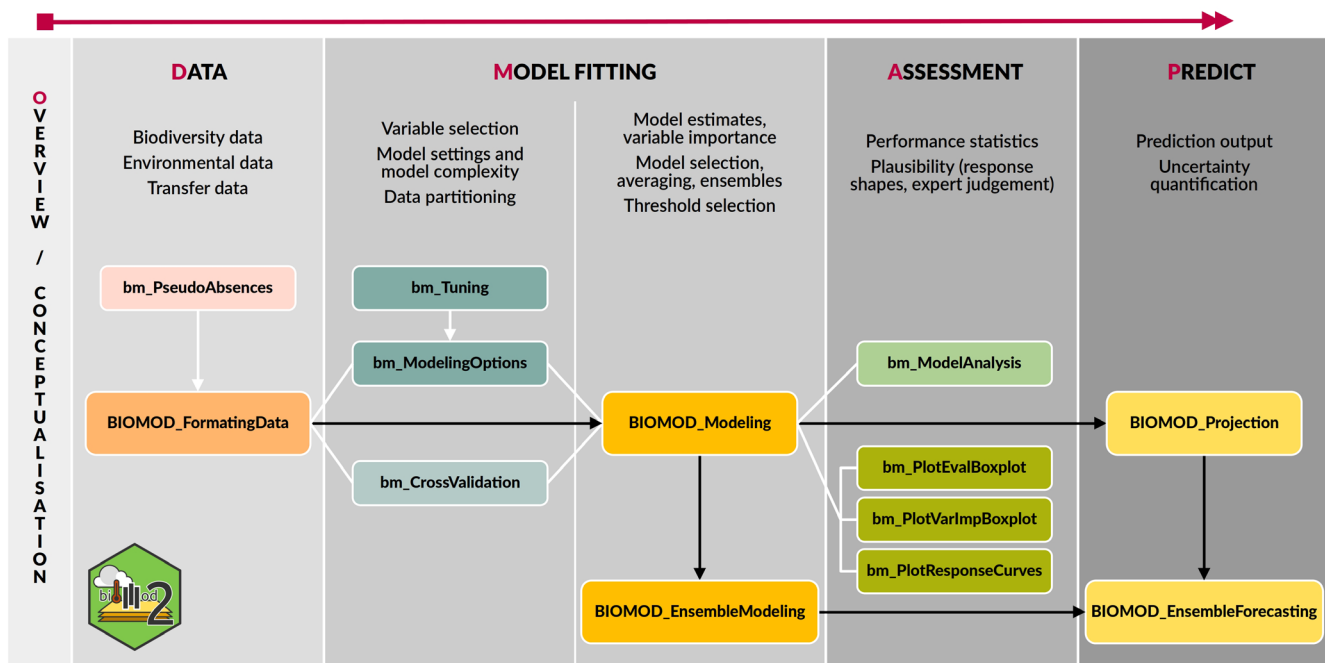


Figure 2. 'biomod2' workflow follows ODMAP protocol steps (Zurell et al. 2020). Main (prefixed *BIOMOD_*[...]) and auxiliary (prefixed *bm_*[...]) functions are highlighted. For a comprehensive list of functions, see the Supporting information.

absence information (the constraints that are implemented in 'biomod2'). Background points, by contrast, make no claim about absence. They simply characterize the environmental space available to the species across the study area, and the model learns to contrast presence locations against this background. They are typically sampled randomly across the landscape, without any assumption about whether the species is absent there or not. The practical consequence is that the two approaches imply different statistical interpretations: pseudo-absence designs aim to estimate the probability of presence (closer to a presence-absence framework), whereas background point designs estimate the relative likelihood of presence compared to the available environment (as in Maxent-style models or point-process models, Renner et al. 2015). In practice, the terminology is not always used consistently in the literature (Guillera-Aroita et al. 2014).

For presence-only datasets (i.e. where genuine absence records are unavailable), 'biomod2' can generate pseudo-absences (PAs) according to several strategies:

- **random**: sample locations where the species has not been recorded – appropriate when presence sampling is non-exhaustive or potentially biased;
- **sre** (surface range envelope): define an environmental envelope encompassing a user-specified proportion of presence-points and sample PAs outside this envelope, under the assumption that the species' environmental niche has been exhaustively sampled;
- **disk**: sample PAs within a ring (disk) around observed presences, excluding points too close (to avoid spatial autocorrelation) or too distant (to respect a localized sampling strategy, Phillips et al. 2009). This approach can also serve as a form of restricted background selection by

limiting sampling to the immediate vicinity of presences; and

- **user.defined**: allow the user to supply a custom set of PA locations.

Regardless of the chosen strategy, PAs are drawn from the given study area, which may be provided either as spatial points (coordinate table) or as a raster layer delineating the study area. A new feature enables the generation of multiple PA sets, each with differing sample sizes, and assignment of those sets to specific algorithms, thereby permitting explicit exploration of how PA sampling effort interacts with algorithmic sensitivity. Additionally, when using *random* or *disk* sampling, the *fact.aggr* parameter can aggregate the study area at coarser resolution to expedite computation without substantially compromising environmental heterogeneity.

Single-model fitting

'biomod2' does not reimplement modelling algorithms itself; rather, it acts as a wrapper that invokes functions from established R packages (Supporting information). The overarching modelling workflow – encompassing data partitioning, algorithm selection, parameter tuning, model fitting, and evaluation – is controlled through the *BIOMOD_Modeling* function. Alternatively, each step may be executed independently using the corresponding auxiliary (prefixed *bm_*[...]) functions.

Cross-validation (*bm_CrossValidation*)

When independent evaluation data are unavailable, 'biomod2' employs data-splitting approaches to partition the dataset into calibration and validation subsets:

- **random**: partition data into 2 subsets, allocating a higher proportion of observations to calibration;
- **kfold**: partition data into k subsets of approximately equal size. Each subset is iteratively used as the validation set while the remaining $k - 1$ subsets serve for calibration. Both **random** and **kfold** splitting can be repeated multiple times (`nb.rep`);
- **block**: following Muscarella et al. (2014), divide the study area into four geographic quadrants (bottom-left, bottom-right, top-left, top-right). This method tests geographical transferability and helps reveal spatial autocorrelation issues;
- **strat** (x-y stratification): based on Wenger and Olden (2012), create $2k$ partitions by stratifying along the x or y coordinate gradients. This approach produces spatially stratified folds that balance sample size and spatial spread;
- **env** (environmental partitioning): for each environmental predictor, generate k partitions subdividing its range into equal-sized bins. This method evaluates model performance across environmental gradients; and
- **user.defined**: allow the user to supply a custom partitioning scheme.

All these methods can be optionally balanced, through a weighting scheme, to ensure an even distribution of presence and absence (or PA) points across folds, mitigating biases arising from spatial clustering.

Modelling options (*bm_ModelingOptions*)

By default, each algorithm's native function arguments are retrieved and exposed to the user. These default parameter settings often reflect generic implementations that may not be optimal for species distribution contexts. To address this, 'biomod2' provides two preset option-sets (Supporting information):

- **bigboss**: provide parameter values fine-tuned by the 'biomod2' development team to better suit typical SDM applications (e.g. appropriate regularization, tree complexity, or learning rates); and
- **tuned**: leverage the 'caret' package to conduct parameter optimization for each algorithm, identifying hyperparameters that improve predictive performance on the user's dataset.

Alternatively, users may specify a `user.defined` list of arguments to override defaults. Options are automatically filtered according to the specified `data.type` (e.g. classification algorithms are used for **binary**, **multiclass** or **ordinal** data, whereas regression algorithms handle **count**, **relative**, or **abundance** data; Supporting information).

Model evaluation (*bm_FindOptimStat*)

To quantify predictive accuracy, 'biomod2' implements a suite of evaluation metrics (`metric.eval`), which are available over single and ensemble models. These metrics are divided into five classes depending on the data type

(Supporting information). For **binary** (presence/absence) and **ordinal** data, most of them are calculated from the confusion matrix, which reports the number of observed presences/absences correctly or incorrectly predicted by the model, also known as true positive/negative (TP, TN) and false positive/negative (FP, FN) values. Common metrics include area under the roc curve (AUCroc), true skill statistic (TSS), Cohen's kappa, and ordinal-specific measures when applicable. For the other data types (**count**, **relative**, and **abundance** data), evaluation relies on continuous-scale metrics such as root mean square error (RMSE), mean absolute error (MAE), Pearson's correlation coefficient, and, for presence-only models, the Boyce index.

In the **binary** context, each metric is first optimized over the *calibration* folds by testing multiple probability thresholds (between 0 and 1) to binarize continuous outputs. This is implemented via the *bm_BinaryTransformation* function. The threshold that maximizes the chosen metric is recorded and then applied to *validation* (and *evaluation*, if independent data are provided) to calculate final performance scores.

Variable importance (*bm_VariablesImportance*)

To assess the relative contribution of each predictor, 'biomod2' employs a permutation-based importance measure analogous to that used in the 'randomForest' package. For a given fitted model:

1. randomly permute the values of the target predictor across all data points;
2. generate model predictions using the permuted dataset;
3. compute Pearson's correlation between the original and the permuted predictions; and
4. repeat steps 1–3 multiple times (`var.import` controls the number of permutations), and average the quantity $(1 - \text{correlation})$ across permutations.

The resulting importance score ranges from 0 (no influence) to 1 (strong influence). Because the permutation is applied independently for each variable, importances are not constrained to sum to one, nor do they capture variable interactions. However, this model-agnostic approach enables direct comparison of variable effects across different algorithms.

Ensemble modelling

Combining multiple single-model predictions into an ensemble prediction can sometimes improve robustness and account for algorithmic uncertainty (Thuiller 2004, Araújo and New 2007, Dormann et al. 2018). Notably, ensemble modelling allows us to test if an average or median prediction is better than single-model ones, but also to explore the variance between models, or extract confidence intervals. In other words, it is not to be taken to generate the ultimate robust predictions, but rather as a way of analysing, comparing, and extracting the best of the different algorithms and sets of cross-validations and pseudo-absence selection.

The *BIOMOD_EnsembleModeling* function supports several aggregation schemes

- *EMmean*: arithmetic mean of continuous predictions across selected models;
- *EMmedian*: median of predictions, reducing the influence of outlier model outputs;
- *EMwmean*: weighted mean, where each model's contribution is proportional to its evaluation score (e.g. TSS);
- *EMci*: confidence intervals (e.g. 95%) across model predictions, enabling spatial assessment of uncertainty;
- *EMcv*: coefficient of variation across model outputs, highlighting regions of high inter-model disagreement;
- *EMca*: committee averaging, converting each model's continuous prediction to a binary map (using its optimized threshold) and computing the proportion of models predicting presence. It represents ensemble's consensus binary prediction;
- *EMmode*: mode of categorical predictions, used for multiclass and ordinal data types and
- *EMfreq*: frequency of the mode of categorical predictions, highlighting regions of high inter-model agreement.

Only single-model runs that meet or exceed a user-specified performance threshold (*metric.select.thresh*) for a given evaluation metric (*metric.select*) are included in the ensemble. Users can also specify the grouping variable (*em.by*) allowing, for example, aggregation across all cross-validation replicates of a single algorithm (*em.by* = 'algo') to explore variability due to data partitioning, or aggregation across different algorithms (*em.by* = 'PA+run') to compare modelling techniques.

Spatial projections

Projections onto novel environmental conditions or scenarios are obtained via the *BIOMOD_Projection* and *BIOMOD_EnsembleForecasting* functions for single-model and ensemble predictions, respectively. When supplying novel predictor rasters, 'biomod2' can automatically identify 'extrapolation' pixels, where one or more environmental variables fall outside the range observed during calibration. The count of out-of-range variables within each pixel is compiled in a 'clamping mask' raster, guiding users to interpret predictions cautiously in regions of environmental novelty. This is analogous to the multivariate environmental similarity surfaces (MESS) analysis which measures the similarity between calibration and projection values for all variables (Elith et al. 2010).

In the case of binary data, continuous (probability) outputs can be transformed to binary maps using the threshold that maximized the chosen evaluation metric (*metric.binary*). Alternatively, users may retain raw probability values and apply a filter (*metric.filter*) to mask out pixels below a specified performance-based cutoff. Binary range maps can be passed to the *BIOMOD_RangeSize* function to compute metrics of range change; specifically, pixel-level gains and losses between two or more scenarios (e.g. current versus future climate projections).

Getters

All primary outputs (e.g. objects of class *BIOMOD.formated.data*, *BIOMOD.models.out*, *BIOMOD.ensemble.models.out*, *BIOMOD.projection.out*) are stored within a structured modelling folder. Auxiliary getter functions (prefixed *get_[...]*) allow users to extract stored components, like species metadata and evaluation datasets; cross-validation partitions and modelling options; performance metrics and variable importance scores; and fitted model objects and corresponding predictions. A summary of available getters is provided in the Supporting information, and the class hierarchy of 'biomod2' objects is illustrated in the Supporting information.

Visualization and plotting

Standard *print* and *plot* functions have been overloaded to accept 'biomod2' object classes (Supporting information). For more customized graphics, auxiliary functions (prefixed *bm_Plot[...]*) accept these objects and produce pre-formatted plots, returning both the underlying data frame and a *ggplot2* object that users can modify further. Available visualizations include:

- boxplots stratified by PA dataset, cross-validation fold, or algorithm for single and ensemble models. They can represent either model evaluations – for calibration, validation, or evaluation partition – or variable importance; and
- response curves following the evaluation strip approach of Elith et al. (2005): i) isolate a single predictor (or a bivariate combination), ii) fix all other variables at a constant (e.g. median), and iii) display predicted changes across the focal predictor's observed range.

Diagnostic analysis and reporting

To interrogate single-model quality and assess modelling assumptions, the *bm_ModelAnalysis* function generates diagnostic plots such as residual heteroscedasticity, outlier detection, and goodness-of-fit visualizations. For comprehensive documentation of an entire modelling workflow, the *BIOMOD_Report* function produces an HTML report containing either:

- reproducible R code snippets used to generate *BIOMOD.formated.data*, *BIOMOD.models.out*, and/or *BIOMOD.ensemble.models.out* objects;
- summaries of data, modelling options, evaluation results, and variable importance; or
- pre-filled ODMAP (overview, data, model, assessment, prediction) protocol tables, following the guidelines of Zurell et al. (2020, <https://odmap.wsl.ch/>), which users can expand or refine as needed (Supporting information).

By integrating these functionalities, 'biomod2' streamlines the end-to-end process of species distribution modelling, from data ingestion, and algorithm selection, through model fitting and validation, to projection, uncertainty quantification, and reporting, while accommodating both traditional presence/absence data and newer abundance-related data types.

Review and discussion

Over the past decade, numerous publications have addressed species distribution modelling (SDM) from both general and specialized perspectives. Broad overviews of SDM methodologies and best practices have been presented by Feng et al. (2019), Zurell et al. (2020), Sillero and Barbosa (2021), Peterson et al. (2022), and Soley-Guardia et al. (2024), while more focused critiques and methodological discussions appear in Descombes et al. (2022), Broussin et al. (2024), and Oeser et al. (2024). In parallel, several reviews have evaluated SDM tools – among them ‘biomod2’ – including Hao et al. (2019), Valavi et al. (2022), and the very recent Kass et al. (2025). The former group of studies (Soley-Guardia et al. 2024) highlight common pitfalls in SDM practice and offer recommendations for mitigating them, whereas the latter (Kass et al. 2025) critique the extent to which existing softwares can address these challenges and advocate for further development.

Addressing common pitfalls (‘Hazards’) in SDM

Soley-Guardia et al. (2024) rightly identified ten hazards that frequently undermine the reliability of SDM outputs. Below, we describe how ‘biomod2’ (ver. 4.3-4) provides tools or workflow options to mitigate most of these hazards, with a few exceptions that remain the responsibility of the modeller rather than the software itself (guidelines in Sillero et al. 2021).

Hazard 1: overlooking errors in occurrence datasets

Data-cleaning – outliers, coordinate errors, taxonomic misidentifications – remains a fundamental step that must be performed prior to modelling. Although ‘biomod2’ provides some filtering (e.g. `filter.raster` in *BIOMOD_FormatingData* to remove duplicate records at a given spatial resolution), detecting erroneous or mislocated occurrences generally falls outside the scope of an SDM wrapper and requires careful scrutiny by the analyst.

Hazard 2: disregarding biases inherent to biological sampling

Presence-only and presence-absence datasets frequently suffer from spatial sampling biases (e.g. over- or underrepresentation of certain regions or habitats). Common correction methods include:

- bias files or target-group background (Ponder et al. 2001, Phillips et al. 2009): sampling PAs in regions with similar survey intensity (e.g. using records of taxonomically or ecologically similar species);
- background thickening (or restricted background; Vollerling et al. 2019): limiting PA sampling to a buffer around presence points to approximate areas where sampling effort is comparable; and
- presence thinning (Veloz 2009, Boria et al. 2014): randomly subsampling occurrences to reduce spatial clustering.

Comparative studies (Fourcade et al. 2014, Vollerling et al. 2019, Barber et al. 2022) demonstrate that the effectiveness

of any correction method depends strongly on (1) underlying bias structure and sample size, (2) choice of performance metric (e.g. AUC versus Boyce), and (3) whether the dataset is real or simulated. Moreover, bias-file and target-group approaches require additional information on survey effort or taxa distributions that may be incomplete or themselves biased. Consequently, current consensus often favors presence thinning, especially with large datasets or minimal sampling metadata, whereas background thickening may be preferable when occurrence records are sparse.

In ‘biomod2’, users can employ:

- `filter.raster` (in *BIOMOD_FormatingData*) to remove duplicate records at the resolution of environmental predictors, retaining only one presence per grid cell; and
- `PA.strategy='disk'` (in *bm_PseudoAbsences*) to implement background thickening, by sampling PAs in a ring around presence points, thereby partially correcting for sampling biases and addressing aspects of Hazard 6 (overly large study regions).

Hazard 3: using spatially or temporally inconsistent environmental variables

Ensuring that predictor layers share the same coordinate reference system, spatial resolution, and temporal timeframe is critical. This responsibility primarily lies with the user: ‘biomod2’ cannot automatically detect or correct inconsistencies in spatial or temporal alignment.

Hazard 4: relying on default SDM settings

Because ‘biomod2’ aggregates roughly twelve distinct algorithms, each with various default parameters, naively using defaults can yield suboptimal or even misleading results. To mitigate this, ver. 4.3-4 implements:

- centralized retrieval of algorithm-specific parameters, exposed to users via *bm_ModelingOptions* function;
- a curated set of *bigboss* presets, providing parameter values deemed more appropriate for typical SDM contexts (e.g. tree complexity in GBM, regularization in GAM/GLM); and
- *bm_Tuning* function, which can evaluate candidate hyperparameter sets, either on the full calibration dataset or within each pseudo-absence/cross-validation fold, and supports automated variable or formula selection to safeguard against overfitting (Hazard 5) and improve predictive performance.

Hazard 5: reducing predictor variables irrespective of their information content

Arbitrary elimination of predictors without assessing their relative contributions can lead to exclusion of ecologically meaningful covariates or retention of noisy variables. The *bm_Tuning* function performs variable selection (e.g. stepwise algorithms for GAM/GLM) or automated formula generation (e.g. for tree-based methods), allowing data-driven inclusion/exclusion of covariates. After model fitting, users can investigate whether the response curves

correspond to expected ecological relationships using the *bm_PlotResponseCurves* function, and refine the models accordingly. This model diagnostic can also be complemented with the importance of variable calculation (through *bm_VariablesImportance* or *BIOMOD_Modeling* directly), which reveals the influence of each single variable on the predictions.

Hazard 6: using an overly large study region

Selecting a study region that far exceeds the species' dispersal capability or ecological range can produce spurious absences simply because certain regions were never surveyed. While delineating the optimal extent depends on the modelling goal (e.g. single- versus multi-species, national-scale assessments) and is ultimately a user-driven decision, 'biomod2' can partially alleviate this hazard by:

- employing `PA.strategy='disk'` to restrict PA sampling to an ecologically plausible buffer around known presences, thereby avoiding sampling in regions beyond the species' reach; and
- allowing custom background rasters that limit the spatial domain to areas of known accessibility or survey coverage.

Nevertheless, defining an appropriate extent must be performed carefully by the modeller, ideally informed by species' natural history, expert opinion, or dispersal models. Depending on the available data and knowledge, using an extent not encompassing the whole species range can instead lead to niche truncation (Guisan et al. 2025). While 'biomod2' does not directly cope with this, complementary external multi-scale or hierarchical workflows (Adde et al. 2023) or 'biomod2'-dependent packages (Mateo et al. 2024) can help address this issue.

Hazard 7: misinterpreting metrics of model performance

A widespread practice among 'biomod2' users is reliance on AUC and/or TSS, even when their data lack true absence records. Both metrics assume the presence-absence paradigm and can be misleading for presence-only data, leading to overestimation of model skill. Version 4.3-4 of 'biomod2' includes presence-only-appropriate metrics such as the Boyce index and MPA, as well as continuous-scale metrics (RMSE, MAE, Pearson's *r*) for count/abundance data. However, the package currently does not provide null-model or randomization-based approaches for assessing the influence of background sampling on performance, which would further aid in distinguishing ecological signal from sampling artefact.

Hazard 8: evaluating models with random splits of the data

Randomly partitioning presence-absence or presence-only data into calibration and validation subsets can produce non-independent folds, especially in the presence of spatial autocorrelation. Although random splitting remains common, spatially or environmentally stratified partitioning typically yields more realistic assessments of model transferability, which are supported by the 'biomod2' package via *bm_CrossValidation* function. These approaches reduce spatial (or environmental) autocorrelation among

folds and better approximate the challenge of extrapolating to novel regions (Hazard 9).

Hazard 9: misunderstanding or overlooking the effects of extrapolation

When projecting SDMs onto novel environmental scenarios (e.g. future climates), pixels whose predictor values fall outside the calibration range produce extrapolative predictions that are inherently uncertain. 'biomod2' can generate a clamping mask (via `build.clamping.mask=TRUE` in *BIOMOD_Projection*), to overlay model predictions and help users identify areas of high extrapolation risk.

Hazard 10: failing to properly rescale or normalize predictors

Many authors emphasize the importance of rescaling (e.g. standardizing continuous predictors to zero mean and unit variance) to ensure numerical stability and comparability across algorithms. 'biomod2' does not automatically standardize predictors; rather, it relies on users to preprocess covariates. Common practice therefore remains to apply external routines prior to using *BIOMOD_FormatingData*.

Current usage patterns of 'biomod2'

Between 2013 and 2019, Hao et al. reviewed 224 of 852 'biomod2'-citing papers, drawing two primary conclusions about the use of the package (ver. 3.3-7): 1) insufficient methodological reporting, with many SDM studies providing minimal detail on parameter choices, cross-validation schemes, or pseudo-absence strategies, hence hindering reproducibility (for similar concerns, see Zurell et al. 2020 and Kellner et al. 2025); 2) over-reliance on default parameters often resulting in poorly tuned models, a phenomenon later corroborated by Valavi et al. (2022).

To assess whether usage patterns have evolved since then, with the considerable acceleration of the package's adoption (Fig. 1), we replicated the previous study by examining a subset of 389 peer-reviewed papers out of 709 published in 2024 and containing the word 'biomod2'. Our preliminary findings (detailed in the Supporting information) echo the 2019 review:

- pseudo-absences and cross-validation: over 60% of studies employed random PA selection and 80% random data splitting, often without justifying these choices in relation to sampling bias or spatial autocorrelation;
- evaluation metrics: a majority (around 70%) reported only AUC and/or TSS – irrespective of whether true absences were available – while presence-only or continuous-data metrics (e.g. Boyce, RMSE) were seldom used;
- algorithm settings: over 60% of papers provided no indication of tuning or parameter values, while 11% relied solely on default parameter values; and
- methodological transparency: fewer than 20% provided code snippets or ODMAP tables; package version information was omitted in over half of the surveyed studies.

These patterns suggest that, despite the increased functionality and documentation of 'biomod2', a large fraction of users continue to utilize it as a 'black box' without

fully exploiting its capacity for rigorous methodological customization. We emphasize that ‘biomod2’ is ultimately a framework: careful model specification (e.g. data preprocessing, parameter tuning, and appropriate validation) remains the user’s responsibility. Although default settings facilitate rapid prototyping, they are not optimized for every ecological context and may yield suboptimal or misleading predictions if uncritically applied.

This was highlighted in Valavi et al. (2022), who conducted a systematic comparison demonstrating that ensemble predictions generated by ‘biomod2’ ver. 3.3-7.1 with the default settings performed no better than simpler modelling approaches, and worse than an externally fine-tuned ensemble model. To evaluate the package enhancements, we replicated their workflow with ‘biomod2’ ver. 4.3-4, running each model with both the default and extended options (bigboss presets). From a predictive performance perspective, we found that 1) fine-tuned parameters tend to improve the results of some single models (DNN, SRE, XGBoost, CTA, and RF), and 2) the newly computed ensemble models perform similarly to those built by Valavi et al. (2022) (detailed in the Supporting information). While this confirms the improved robustness of the package in its most common use, we encourage users to fully explore modelling choices and algorithm settings wherever possible.

Supplementary to online documentation and examples, we provide here a case study detailing a complete modelling workflow with ‘biomod2’ ver. 4.3-4 and presenting the different steps and functions (Supporting information). It includes a Material and methods template designed to help users reporting their approach when using the package. Regarding citation practices, many authors continue to reference early descriptions (Thuiller 2003, 2014, Thuiller et al. 2009) and rarely the exact version used. With this paper, we hope to improve this lack of a clear reporting framework already highlighted by many authors (Zurell et al. 2020, Kellner et al. 2025) by providing a concise methodology and ensuring that essential details, including package version and citation, are systematically and explicitly documented (Supporting information).

Concluding remarks

‘biomod2’ ver. 4.3-4 represents a substantial advance in the SDM toolbox, notably by:

- providing a cohesive structure, through main (prefixed *BIOMOD_* [...]) and auxiliary (prefixed *bm_* [...]) functions, that encourages transparent workflows and facilitates reproducibility;
- supporting non-binary data types (counts, multiclass, ordinal, relative and absolute abundance values) alongside traditional presence-absence and presence-only frameworks;
- allowing flexible parameterization and automated tuning (via *bm_ModelingOptions* and *bm_Tuning*),

which help mitigate the tendency to rely on default settings;

- enabling robust cross-validation and extrapolation diagnostics, thereby addressing many of the hazards identified by Soley-Guardia et al. (2024); and
- integrating comprehensive reporting tools, including pre-filled ODMAP tables, to foster more rigorous methodological documentation.

Nevertheless, the utility of any SDM framework ultimately hinges on careful user stewardship. Data quality control (Hazard 1) and study extent delineation (Hazard 3) remain critical tasks that must be performed prior to invoking ‘biomod2’. Choice of pseudo-absence strategies (Hazard 2) and background extent (Hazard 6) must reflect the species’ ecology, sampling effort, and research goals. Appropriate metric selection (Hazard 7) and robust validation schemes (Hazard 8) require user awareness and should match data type and study objectives.

In conclusion, while ‘biomod2’ ver. 4.3-4 equips ecologists with a powerful, flexible, and transparent SDM platform, its effectiveness depends on conscientious application: careful data curation, appropriate algorithm tuning, and rigorous validation. By combining these best practices with the expanded functionalities of ‘biomod2’, researchers can produce more reliable and ecologically meaningful distribution models in both presence-absence and abundance contexts.

To cite ‘biomod2’ or acknowledge its use, cite this Software note as follows, substituting the version of the application that you used for ‘version 1.0’:

Guéguen, M., Blancheteau, H., Lemaire-Patin, R. and Thuiller, W. 2026. ‘biomod2’ – extending presence-absence species distribution models to multiple data types. – *Ecography* 2026: e08588 (ver. 1.0).

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Author contributions

Maya Guéguen: Conceptualization, Investigation, Methodology, Software, Supervision, Visualization, Writing - original draft, Writing - review & editing. **Hélène Blancheteau:** Software, Methodology, Validation, Writing - review & editing. **Rémi Lemaire-Patin:** Software,

Validation, Writing - review & editing. **Wilfried Thuiller:** Conceptualization, Investigation, Methodology, Supervision, Funding acquisition, Writing - original draft, Writing - review & editing.

Data availability statement

Data are available from the Zenodo Repository: Appendix A – <https://doi.org/10.5281/zenodo.17831076>; Appendix B – <https://doi.org/10.5281/zenodo.17898629>; Appendix C – <https://doi.org/10.5281/zenodo.17898708> (Guéguen et al. 2026).

Supporting information

The Supporting information associated with this article is available with the online version.

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